



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

AI-Powered Insurance Claim Automation System

**Generative AI
CA - 5**

Submitted By:

**Shrishti Prasad & Raghav Uparkar
22070521035 & 22070521045
VII A**

Submitted To:

**Dr. Nisha Balani
Dr. Poorva Agrawal**

**Department of Computer Science and Engineering
Symbiosis Institute of Technology, Nagpur**



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Department of Computer Science and Engineering

AI-Powered Insurance Claim Automation System

Abstract:

In a conventional manner, the verification of insurance claims is manual, lengthy, and difficult to handle through human intervention thus causing bias and mistakes. Thousands of documents like claims forms, bills of repairs, surveyors' reports, and insurance policy histories must be scrutinized by the insurers before making a final decision of either approving or rejecting a claim. This project, however, has presented a solution to this problem by an AI-based automated system that not only sorts the uploaded insurance documents but also predicts the claim's decision, and gives human-friendly justifications in support of the ruling.

The solution is built on three fundamental pillars: a Document Classification Agent, a Claim Decision Prediction Model, and an AI Explanation Generator. The classification agent applies Natural Language Processing (NLP) methods together with a zero-shot classification model to determine the document's type based on the content of the text. The prediction model employs a machine learning method (Random Forest Classifier) to extract and evaluate policy and vehicle-related characteristics, and then he decides if the claim ought to be accepted or not. An explanatory generation model is then used to offer reasoning that is easy to understand and concise for the decision making process.

By this whole procedure, the insurance companies have less to do manually, the process of claim clearing becomes faster, there is more clarity and less cost of operation. Therefore, AI can be seen as a major factor in the future digital transformation of many organizational processes.

1. Introduction:

Insurance claim management is the very foundation of the insurance business. When a client makes a claim due to a vehicular accident, the insurance company has to check different papers, ascertain the claim's genuineness, and decide if the claim is in line with the policy. This task is usually done by human agents who are slow, inconsistent and costly in their operations.

The automation and AI technologies have been growing at a rapid pace to such an extent that they are now viewed as a way of overcoming the mentioned issues. Supervised learning models can be trained on historical claims data to predict future claim outcomes just as transformer-based language models (e.g. BERT) have made document interpretation hi-tech, precise, and context-aware in a quicker and more



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Department of Computer Science and Engineering

reliable manner. Natural Language Processing and machine learning, especially AI, are capable of accomplishing this task by analyzing documents, finding patterns, and even supporting decision making much quicker and more accurately.

The aim of this work is the establishment of an AI automation framework consisting of three main phases for processing insurance claims. The first phase is the automatic document classification in which, based on reading its text content, the system detects what type of the document has been uploaded. The second phase is prediction of claim decision in which the model takes into account vehicle and policy features determining if the claim is likely to be accepted or declined. The third phase involves the generation of a human-friendly explanation of the decision thus, the automated system would maintain transparency and trust.

The goal of the project will be to lighten the manual burden on the staff, speed up the processing, decrease errors, and aid the right decision in the field of insurance claim assessment.

2. Methodology:

- **Tools & Technologies Used:** Python, Streamlit, OpenAI API

Tools and technologies used Component	Technology Used
Programming Language	Python
Document Text Extraction	pdfplumber/OCR
NLP Model	Zero-shot BERT Model
ML Model	Random Forest Classifier
Libraries	Pandas, Sklearn, Transformers
Interface/Execution	Google Colab , Streamlit

Process:

1. The user uploads insurance-related documents (PDF).



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

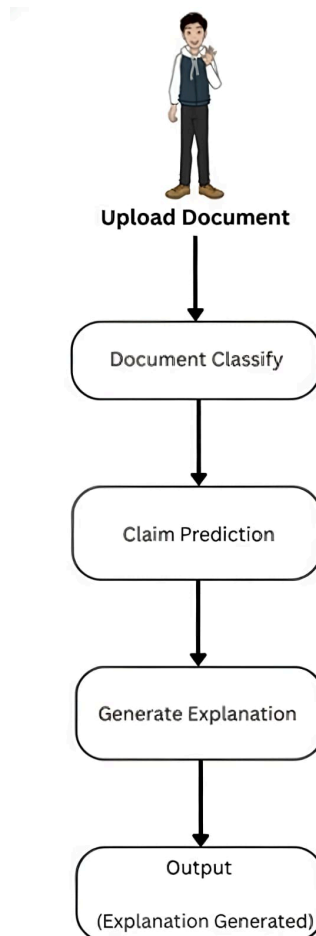
Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Department of Computer Science and Engineering

2. Text is extracted and passed to the **Document Classification Model**.
3. The document category is predicted with a confidence score.
4. Policy and vehicle-specific features are input to the **Random Forest Model**.
5. The system predicts whether the claim is **Approved** or **Denied**.
6. The Explanation Module generates a natural language justification.

- **Method/Pseudocode**





SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Department of Computer Science and Engineering

Step 1: Input and Text Extraction

The process begins with the user uploading an insurance-related document (such as an invoice, claim form, or inspection report).

- The system extracts the text content from the uploaded PDF using pdfplumber or Tesseract OCR (in case of scanned images).
- The extracted text is cleaned and preprocessed to remove unnecessary spaces, symbols, and formatting noise.

Step 2: Document Classification (Zero-Shot Classification)

The preprocessed text is passed to a Zero-Shot BERT Model from the Transformers library.

- This model classifies the document into predefined categories such as *Invoice*, *Claim Form*, or *Report* without the need for additional training data.
- The model outputs the most probable document type with a confidence score. This step helps the system automatically understand what type of document has been uploaded.

Step 3: Claim Decision Prediction (Random Forest Model)

The system takes structured claim-related inputs - such as vehicle age, safety rating (NCAP), presence of safety features, and customer details.

- These features are passed into a Random Forest Classifier trained on historical claim data.
- The model predicts whether the claim should be Approved or Denied based on the identified risk patterns.

Step 4: AI Explanation Generation

After prediction, the AI Explanation Bot generates a human-like explanation for the result:

- If Approved, it highlights positive factors like good safety ratings and eligibility.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Department of Computer Science and Engineering

- If Denied, it explains reasons like high risk or lack of safety features. The explanation is created using NLP techniques and predefined reasoning logic, ensuring transparency and interpretability.

Step 5: Output

Finally, the system displays the combined output:

- Document Type
- Claim Decision (Approved / Denied)
- AI-Generated Explanation

If Approved:

Generate Explanation on Safety & Policy Eligibility

Else:

Generate Explanation on Risk & Non-Compliance

Output: Document Type + Prediction + Explanation

3. Results with Screenshots:

- Document Classification Accuracy ~ 90%+
- Claim Decision Prediction Confidence > 85%
- Explanation Generation clearly communicates the reasoning



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

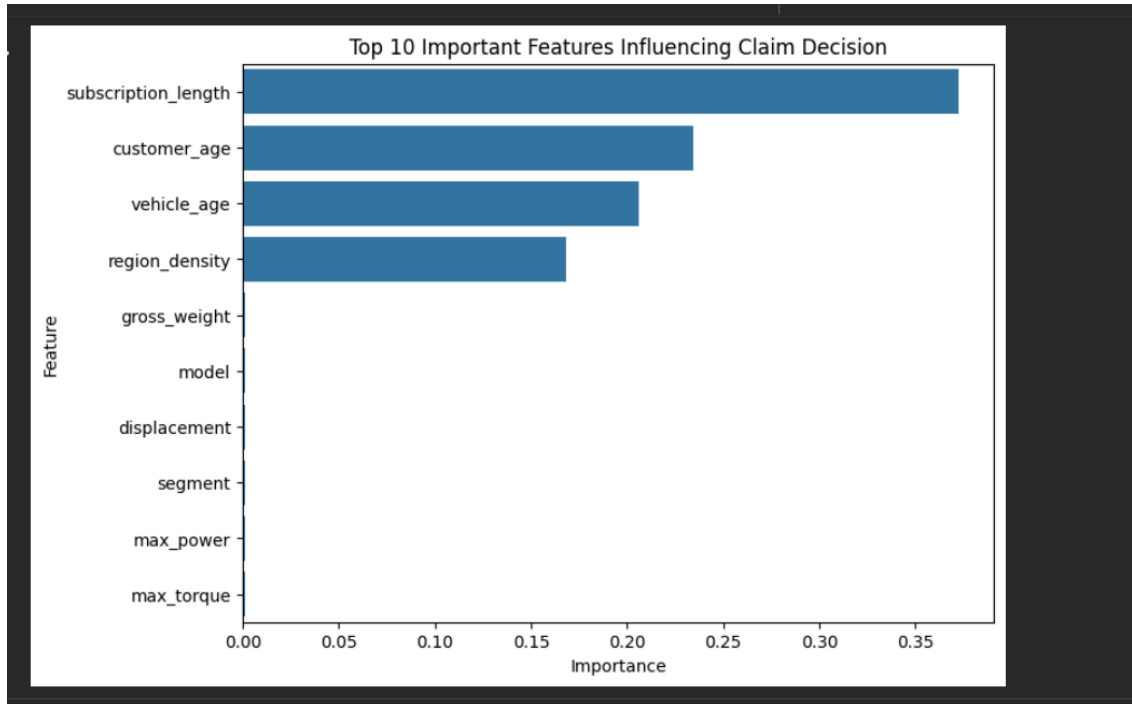
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Department of Computer Science and Engineering



```
[17] ✓ 10s
print("\n✅ XGBoost Model Trained and Evaluated:")
print("\n🔍 Accuracy (XGBoost):", round(accuracy_score(y_test, y_pred_xgb)*100, 2), "%")
print("\n📊 Classification Report (XGBoost):\n", classification_report(y_test, y_pred_xgb))

... ✅ Random Forest Model Trained and Evaluated:

🔍 Accuracy (Random Forest): 98.62 %

📊 Classification Report (Random Forest):
      precision    recall  f1-score   support

    0.0         1.00      0.97      0.99         5733
    1.0         0.97      1.00      0.99         5972

 accuracy          0.99
 macro avg          0.99
weighted avg          0.99

✅ XGBoost Model Trained and Evaluated:

🔍 Accuracy (XGBoost): 79.01 %

📊 Classification Report (XGBoost):
      precision    recall  f1-score   support

    0.0         0.85      0.70      0.76         5733
    1.0         0.75      0.88      0.81         5972

 accuracy          0.79
 macro avg          0.80
weighted avg          0.80
```



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Department of Computer Science and Engineering

```
else:
    print("✗ Could not extract text from the PDF.")
else:
    print("No file uploaded.")

... Upload a PDF for classification:
    Choose Files icf.pdf
    icf.pdf(application/pdf) - 212231 bytes, last modified: 11/7/2025 - 100% done
    Saving icf.pdf to icf.pdf
    ✓ Uploaded: icf.pdf

    --- Extracted Text Preview ---
    INSURANCE CLAIM FORM
    Policy No: PL-45321
    Accident Date: 12 June 1975
    Name: Suresh
    Age: 50

    -----

    Document split into 1 chunk(s).
    Running classification...
```

📄

Enter claim details below:

Vehicle Age (in years): 15

Customer Age: 50

Fuel Type (Petrol/Diesel/CNG): petrol

Vehicle Segment (A1/A2/B1/C1 etc.): a1

NCAP Rating (1-5): 2

Brake Assist (Yes/No): yes

Power Steering (Yes/No): no

Central Locking (Yes/No):



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Department of Computer Science and Engineering

Enter claim details below:

Vehicle Age (in years): 15
Customer Age: 50
Fuel Type (Petrol/Diesel/CNG): petrol
Vehicle Segment (A1/A2/B1/C1 etc.): a1
NCAP Rating (1-5): 2
Brake Assist (Yes/No): yes
Power Steering (Yes/No): no
Central Locking (Yes/No): no

Predicted Claim Decision: **✗ DENIED**
Model Confidence: 13.0%

Generative AI Explanation:
The customer's age is 50 years and the vehicle has a 15.0 year old age. The vehicle has a 2 NCAP rating and is a petrol vehicle. The customer's age is 50.

Detailed Explanation:
The vehicle is older, which often increases the risk of component failures or wear-and-tear claims.
An NCAP rating of 2 shows moderate safety performance.
The vehicle includes modern safety features such as brake assist, emergency warning system, speed alert system, which enhances driver safety.
The customer's age is within a low-risk range (25-60 years).

✗ The claim is likely DENIED due to a combination of older vehicle age, lower safety rating, or limited safety systems.

Detailed Explanation:
The vehicle is older, which often increases the risk of component failures or wear-and-tear claims.
An NCAP rating of 2 shows moderate safety performance.
The vehicle includes modern safety features such as brake assist, emergency warning system, speed alert system, which enhances driver safety.
The customer's age is within a low-risk range (25-60 years).

✗ The claim is likely DENIED due to a combination of older vehicle age, lower safety rating, or limited safety systems.

AI-Powered Insurance Claim Automation System

Modules:

- Document Classification Agent - Detects the document type from uploaded PDF
- Claim Decision Predictor - Predicts whether a claim is Approved **✓** or Denied **✗**
- Generative Explanation Bot - Explains the decision in simple language

Step 1: Upload a Document (PDF)

Upload PDF (Claim Form / Invoice / Report / etc.)

Drag and drop file here
Limit 200MB per file • PDF

icf.pdf 207.3KB

Text extracted successfully!

Extracted Text Preview



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Department of Computer Science and Engineering

35

-

+

Fuel Type

Petrol

▼

Brake Assist

Yes

No

Central Locking

Yes

No

Predict Claim Decision

Prediction Result: ✗ Denied

Explanation

The customer's age is 35 and the vehicle is 15.0 years old. It has brake assist, power steering, central locking and a NCAP rating of 4. The insurance claim decision is Denied.

Developed by Shrishti & Raghav | B.Tech CSE | AI-Powered Insurance Claim Automation System

Predicted Document Type: Claim Form (Confidence: 84.96%)

Step 2: Predict Claim Approval / Denial

Vehicle Age (years)

15.00

-

+

Customer Age

35

-

+

Fuel Type

Petrol

▼

Vehicle Segment

A1

A1

A2

B1

C1

C2

Power Steering

Yes

No

Central Locking

Yes

No

Predict Claim Decision

Prediction Result: ✗ Denied



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Department of Computer Science and Engineering

35 - +

Fuel Type: Petrol

Brake Assist: Yes

Central Locking: Yes

Predict Claim Decision

Prediction Result: ❌ Denied

Explanation

The customer's age is 35 and the vehicle is 15.0 years old. It has brake assist, power steering, central locking and a NCAP rating of 4. The insurance claim decision is Denied.

Developed by Shrishti & Raghav | B.Tech CSE | AI-Powered Insurance Claim Automation System

4. Applications:

- Insurance Companies (Vehicle Claims) - The system can be used by vehicle insurance companies to automate claim verification. It can read uploaded documents such as invoices, claim forms, and inspection reports, predict whether a claim is valid, and generate clear explanations for approval or rejection — saving time and reducing human bias.
- Health & Property Claim Processing - In health or property insurance, large numbers of claims are submitted with bills, medical records, or damage reports. The AI model can automatically classify these documents, extract important details, and help insurers make faster and more accurate claim decisions while maintaining transparency.
- Government Subsidy Document Verification - Government welfare and subsidy programs often require verification of beneficiary documents like income proofs, IDs, and certificates. The system can automate the validation process, detect incomplete or invalid submissions, and speed up approval of genuine applications.
- Banking Loan Document Screening - Banks receive numerous applications for personal and business loans.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Department of Computer Science and Engineering

The AI system can analyze loan documents, verify applicant information, and predict approval likelihood based on predefined rules. This helps financial institutions automate loan processing, ensure compliance, and minimize fraud risk.

5. Future Scope:

- Deploy as full Web Application / Mobile App
- Add image-based damage assessment model (CV + Deep Learning)
- Multi-language document understanding
- Integration with real-time insurance databases

6. Conclusion:

The proposed AI-Powered Insurance Claim Automation System presents an intelligent and efficient solution to automate the insurance claim evaluation process by leveraging Machine Learning and Natural Language Processing (NLP) techniques. Traditional claim processing methods are manual, time-consuming, and prone to human error. This system addresses these challenges by integrating three core components - Document Classification Agent, Claim Decision Predictor, and AI Explanation Bot - into a unified automation pipeline.

The Document Classification Agent automatically identifies the type of uploaded document, such as an invoice or claim form, using Zero-Shot Classification based on a pre-trained BERT model. The Claim Decision Predictor, developed using the Random Forest algorithm, analyzes policy features such as vehicle age, safety rating, and customer details to predict whether a claim should be approved or denied. Finally, the AI Explanation Bot generates a human-understandable justification for each decision, enhancing transparency and user trust.

Through these components, the system minimizes manual intervention, reduces claim processing time, and ensures consistent and data-driven decisions. Moreover, by providing explainable AI outputs, it promotes accountability and confidence among policyholders and insurers alike.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Department of Computer Science and Engineering

This project successfully demonstrates how Artificial Intelligence can revolutionize traditional insurance workflows by automating repetitive tasks and improving operational efficiency. The approach establishes a strong foundation for future enterprise-level deployment and further enhancements, such as integrating image-based claim analysis and real-time fraud detection mechanisms.

References:

https://colab.research.google.com/drive/1YAKCMUseV6Vmi4EYrvUOTandm_c7miDS#scrollTo=6287fc5c